

Step 1: Open Terminal Press `Cmd + Space` to open Spotlight, type **Terminal**, and press `Return`.

Step 2: Install Developer Command Line Tools (C Compiler) Instead of installing WSL, run this single command in Terminal to install the C compiler toolchain:

```
medikondarakesh@Medikondas-MacBook-Air ~ % xcode-select --install
xcode-select: note: Command line tools are already installed. Use "Software Update" in System Settings or the softwareupdate command line interface to install updates
medikondarakesh@Medikondas-MacBook-Air ~ %
```

Step 3: Verify the Compiler Installation Check that your C compiler is installed and ready:

```
medikondarakesh@Medikondas-MacBook-Air ~ % gcc --version
Apple clang version 21.0.0 (clang-2100.1.1.101)
Target: arm64-apple-darwin25.5.0
Thread model: posix
InstalledDir: /Applications/Xcode.app/Contents/Developer/Toolchains/XcodeDefault.xctoolchain/usr/bin
medikondarakesh@Medikondas-MacBook-Air ~ %
```

Step 4: Create Your C Program Create a new file named `process.c` using the command-line text editor:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed.\n");
        return 1;
    } else if (pid == 0) {
        // Child process code execution
        printf("Child Process [PID: %d | Parent PID: %d] running...\n", getpid(), getppid());
    } else {
        // Parent process execution
        printf("Parent Process [PID: %d | Child PID: %d] waiting...\n", getpid(), pid);
    }

    return 0;
}
```

Step 5: Compile the Program Compile your `process.c` file using the standard C compiler

```
medikondarakesh@Medikondas-MacBook-Air: ~/Applications/Xcode.app/Contents/Developer/Toolchains/XcodeDefaultToolchain/bin$  
medikondarakesh@Medikondas-MacBook-Air ~ % nano process.c  
medikondarakesh@Medikondas-MacBook-Air ~ % gcc process.c -o process  
medikondarakesh@Medikondas-MacBook-Air ~ %
```

Step 6: Run the Program Execute the compiled binary directly in your terminal:

```
medikondarakesh@Medikondas-MacBook-Air ~ % gcc process.c -o process  
medikondarakesh@Medikondas-MacBook-Air ~ % ./process  
Parent Process [PID: 57058 | Child PID: 57059] waiting...  
Child Process [PID: 57059 | Parent PID: 57058] running...  
medikondarakesh@Medikondas-MacBook-Air ~ %
```

Step 7: Trace Process Execution (Mac Alternative to `strace`)

```
[medikondarakesh@Medikondas-MacBook-Air ~ % sudo dtruss ./process
>Password:
dtrace: system integrity protection is on, some features will not be available

SYSCALL(args)          = return
Parent Process [PID: 57079 | Child PID: 57080] waiting...
Child Process [PID: 57080 | Parent PID: 57079] running...

medikondarakesh@Medikondas-MacBook-Air ~ % █
```

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 1 Branch

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